Survey-Weighted and Temporally Validated Risk Modeling for Under-Five Mortality Using Bangladesh DHS Data

Complete study plan

1. Gap and aim

Existing Bangladesh U5M models report inflated accuracy (random splits, SMOTE, no survey weights, leakage), or validate temporally without full survey weighting and with no division-level policy output. Aim: measure how validation design and survey weighting change results, then deliver a division-level risk and screening scenario for policy.

2. Gaps in the four anchor papers

Paper	Main gaps
Naznin 2025 (U5M, PLOS One)	Boruta run on full data before split, SMOTE with unclear test handling, random 70/30 split, no survey weights, no calibration, post-outcome predictors (breastfeeding, children ever born), 44% complete-case loss.
Fahim 2026 (U5M, arXiv)	Single temporal cut, weights only as sensitivity check, mixed neonatal and later deaths, censoring, non-algorithmic feature selection, borderline division result (n=8), no CIs on NNS.
Haque 2026 (stunting)	No survey weights, undersampling, feature set chosen on hold-out test data, weak calibration.
Chandra 2026 (hypertension)	Weights only in one model, no formal feature selection, large train to test gap (overfitting).

3. Research questions

1. How much does temporal versus random validation change performance and screening workload?
2. How much do survey weights change prevalence, calibration and division rankings?
3. Is SWIFT-U5 better calibrated and more stable across rounds than standard models?
4. Which divisions carry the highest risk and best screening yield, and which drivers are shifting?

4. Data

- BDHS BR files: 2004, 2007, 2011, 2014, 2017-18, 2022.
- Births in the 5 years before each survey, exposure-complete only.
- Weights (v005), PSU (v021), strata (v022).
- Division harmonization: 7 divisions for 2004 to 2014 trends (Mymensingh did not exist), 8 divisions for 2017-18 and 2022.

5. Predictors

Tiered by timing. **T0 structural:** maternal age at birth, education, wealth, residence, division, birth order, birth interval, parity before. **T1 antenatal:** ANC visits, TT, iron. **T2 birth:** facility delivery, caesarean, birth size. Excluded as post-outcome: breastfeeding, child age, children ever born, current vaccination. Train-only imputation plus missingness flags.

6. Feature selection (training data only)

1. Domain filter: pre-outcome, comparable across rounds, under 30% missing, drop one of any pair with correlation above 0.8.
2. Weighted LASSO logistic plus permutation importance on weighted XGBoost, inside PSU-grouped folds.
3. Stability selection: keep features chosen in 70% or more of runs by both methods.
4. Repeat per rolling window to label features stable or drifting.

5 SHAP only for interpretation after the model is fixed, never for selection.

7. The 2 x 2 design

Arm	Validation	Weights
A	Random split	No
B	Random split	Yes
C	Forward-chained temporal	No
D	Forward-chained temporal	Yes

- Plus a matched-n random arm and a random-split result scored on the test round only, to separate validation effect from test-set composition.
- Same features, models and tuning in every arm. No SMOTE. Class weights multiplied by survey weights, normalized within round.
- Forward chaining: 2004 to 2007, 2004-07 to 2011, and so on to 2022.
- Mixed check: train unweighted and evaluate weighted, and the reverse, to locate where weight damage arises.

8. SWIFT-U5 framework

Survey-Weighted, Importance-corrected, Forward-validated Temporal risk model for U5M.

- **Module A, recency and shift weighting:** survey weight x time decay x clipped density ratio from a train versus next-round domain classifier. Decay rate tuned by forward chaining.
- **Module B, division-shrunk head:** division intercept and slope on the logit, shrunk toward national values by empirical Bayes in proportion to death counts.
- **Module C, drift-aware recalibration:** calibration slope and intercept tracked per round and division, recalibrated on the latest round, plus a SHAP drift index.
- Baselines: weighted logistic, XGBoost, random forest. Ablation removes each module.

9. Evaluation

Weighted and unweighted AUROC, AUPRC, Brier score, calibration slope and intercept, sensitivity, PPV and NNS at top 5, 10 and 20 percent capacity, decision curve net benefit. PSU-stratified cluster bootstrap CIs and paired bootstrap for differences between arms.

10. Division policy layer

- Risk atlas and trend by division.
- Division NNS and capacity table with CIs.
- Top drivers per division from SHAP.
- Equity audit by wealth quintile and residence.
- Priority table: expected deaths captured per 1,000 births screened.

11. Results structure

#	Item	Content
1	Table 1	Weighted sample, deaths, prevalence by round and division
2	Table 2	2 x 2 main result with CIs
3	Fig 1	Inflation plot, random versus temporal, across six rounds
4	Table 3	Effect decomposition: validation effect, weight effect, interaction
5	Fig 2	Calibration curves, arms A to D
6	Fig 3	Feature selection frequency, stable versus drifting

#	Item	Content
7	Table 4	SWIFT-U5 versus baselines per test round
8	Table 5	Ablation of modules
9	Fig 4	Division risk atlas and trend
10	Table 6	Division NNS and capacity
11	Fig 5	SHAP drift by division
12	Table 7	Equity audit

12. Claimed contributions

1. First controlled factorial assessment of validation design and survey weighting for Bangladesh U5M prediction.
2. Six-round, exposure-complete, leakage-controlled forward-chained benchmark.
3. SWIFT-U5, an integrated survey-aware, drift-aware framework (a framework, not new theory).
4. Division-level screening estimates with uncertainty, plus temporal feature stability.

Not new, do not claim: recency weighting, density ratio correction, empirical Bayes shrinkage, Platt recalibration, SHAP, stability selection, temporal validation itself.

13. Scope and risks

- **ICCIT version:** under-5 outcome, 2 x 2 design, feature selection, Modules A and C, division NNS table, SHAP drift.
- **Journal version:** add neonatal and infant windows, Module B, full equity audit.
- Expect AUROC near 0.70 to 0.75. The story is calibration, stability and division usefulness.
- Check death counts per division per round first. Drop any module that shows no gain in a pilot.
- Variable harmonization across 2004 to 2022 and missing data losses must be reported.

14. Order of work

1. Harmonize variables and build the leakage-free dataset.
2. Run the 2 x 2 with baselines (core result).
3. Feature selection and stability.
4. SWIFT-U5 modules and ablation.
5. Division layer and figures.
6. Write.